

Jianjin Xu

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EDUCATION

Carnegie Mellon University

Ph.D. in Robotics, Advisor: Prof. Fernando De la Torre Frade

Pittsburgh, PA

8/2023 –

Columbia University

M.S. in Computer Science, Advisor: Prof. Changxi Zheng

New York, NY

8/2019 – 5/2021

Tsinghua University

B.Eng. in Computer Science, Advisor: Xiaolin Hu

Beijing, CN

8/2015 – 7/2019

RESEARCH INTERESTS

3D Human Avatar Generation, Video Diffusion Model for Human, Generative Model, 3D Computer Vision

PUBLICATIONS AND MANUSCRIPTS

1. **Jianjin Xu**, Soham Dasgupta, Fernando De la Torre. ARG-Avatar: Animation-Rendering Guidance for Lightweight Feedforward Avatar Reconstruction. (Under Review at 3DV 2027)
2. Li, Hengjia and **Jianjin Xu** and Cheng, Keli and Wang, Lei and Bi, Ning and Wu, Boxi and De la Torre, Fernando and Cai, Deng. [Multi-View Consistent Human Image Customization via In-Context Learning](#). (ArXiv, 2025)
3. Lin, Yukang and Fung, Hokit and **Jianjin Xu** and Ren, Zeping and Lau, Adela SM and Yin, Guosheng and Li, Xiu. [Mvportrait: Text-guided motion and emotion control for multi-view vivid portrait animation](#). (CVPR 2025)
4. Lin, Yuchen and Lin, Chenguo and **Jianjin Xu** and MU, Yadong. [OmniPhysGS: 3D Constitutive Gaussians for General Physics-Based Dynamics Generation](#). (ICLR 2025)
5. Chen, Anthony and **Jianjin Xu** and Zheng, Wenzhao and Dai, Gaole and Wang, Yida and Zhang, Renrui and Wang, Haofan and Zhang, Shanghang. [Training-free regional prompting for diffusion transformers](#). (ArXiv, 2024)
6. Kwon, Youngjoong and Fang, Baole and Lu, Yixing and Dong, Haoye and Zhang, Cheng and Carrasco, Francisco Vicente and Mosella-Montoro, Albert and **Jianjin Xu** and Takagi, Shingo and Kim, Daeil and others. [Generalizable human gaussians for sparse view synthesis](#). (ECCV 2024)
7. **Jianjin Xu**, Saman Motamed, Praneetha Vaddamanu, Chen Henry Wu, Christian Haene, Jean-Charles Bazin, Fernando de la Torre. [Personalized Face Inpainting with Diffusion Models by Parallel Visual Attention](#). (WACV 2024)
8. Saman Motamed, **Jianjin Xu**, Chen Henry Wu, Fernando De la Torre. [PATMAT: Person Aware Tuning of Mask-Aware Transformer for Face Inpainting](#). (ICCV 2023)
9. **Jianjin Xu**, Zhaoxiang Zhang, Xiaolin Hu. [Extracting Semantic Knowledge from GANs with Unsupervised Learning](#). (TPAMI, 2023)
10. **Jianjin Xu**, Changxi Zheng. [Linear Semantics in Generative Adversarial Networks](#). (CVPR 2021)
11. **Jianjin Xu**, Zheyang Xiong, Xiaolin Hu. [Frame Difference-Based Temporal Loss for Video Stylization](#). (ArXiv)

PROFESSIONAL EXPERIENCE

Unrestricted Vehicle Re-Identification System with Deep Metric Learning

6/2018 – 10/2018

Internship at MSRA, supervised by Lead Researcher Xun Guo

- Developed a re-identification system that inputs raw videos of monitors and identifies re-appeared vehicles. The system first detects vehicles by faster RCNN, then conducts tracking and matching by learned deep metrics.
- Trained the deep metric model on VeRi dataset and validated it on VID dataset and collected traffic videos.

AWARDS

3rd Prize in 36th the Challenge Cup Competition, Tsinghua University

4/2018

2nd Prize in Mathematical Contest in Modeling, 2017

2/2017

TEACHING EXPERIENCE

TA @ Carnegie Mellon University, 16-822: Geometry-based Methods in Vision, Fall 2024 & 2025

TA @ Columbia University, COMS-W4995 Special Topics In Computer Science, I: Causal Inference, 2020

MISCELLANEOUS EXPERIENCE

Chairman of Tsinghua Future Internet and Computation Club

6/2018-6/2019

SKILLS

pytorch / tensorflow / python / C++ / javascript / CUDA